Modern Academy
for Engineering and Technology in Maadi
Basic Science Department

MODERN ACADEMY FOR ENGINEERING & TECHNOLOGY

Academic Year	2023-2024
Semester	First
Exam Date	14/11/2023
Academic Level:	Sophomore

QUESTIONS FOR THE MID TERM WRITTEN EXAM

Subject: (MTHn103) **Mathematics-3** **Spec.:** Elec. and Mech.
Differential Equations and Transforms
Examiners: Assoc. Prof. Ashraf Taha and Dr. Amira Salem **Time:** 60 minutes

Number of Pages: 1 Number of Questions: 2 Attempt all questions

MODEL ANSWER

Question 1: [LO's: LO 1, LO 2, LO 4] **(10 points)**

Find the general solution of the first ordinary differential equations:

a) $y\dfrac{dy}{dx}=\sec y^2\cos x+\dfrac{x^3}{\cos y^2}$. **(2 points)**

$$y\frac{dy}{dx}=\sec y^2\cos x+x^3\sec y^2=\sec y^2\left(\cos x+x^3\right).$$

$$\int y\cos y^2dy=\int\left(\cos x+x^3\right)dx\rightarrow\boxed{\frac{1}{2}\sin y^2=\sin x+\frac{x^4}{4}+C}$$

b) $xy'-y=x^2\cos^2\left(\dfrac{y}{x}\right)$. **(3 points)**

$$y'-\frac{y}{x}=x\cos^2\left(\frac{y}{x}\right)\rightarrow\frac{y}{x}=u\rightarrow y=xu\rightarrow y'=xu'+u$$

$$xu'+\cancel{u}-\cancel{u}=x\cos^2(u)\rightarrow\int\sec^2(u)du=\int dx$$

$$\tan(u)=x+C\rightarrow\boxed{\tan\left(\frac{y}{x}\right)=x+C}$$

c) $\left(2x\ln y+\sin x\right)dx+\left(\dfrac{\ln y}{y}+\dfrac{x^2}{y}\right)dy=0$. **(2 points)**

$$M_y=\frac{2x}{y}=N_x\left(Exact\ D.E\right)\rightarrow\int\left(2x\ln y+\sin x\right)dx+\int\left(\frac{\ln y}{y}+\frac{x^2}{y}\right)dy=C$$

$$x^2\ln y-\cos x+\frac{(\ln y)^2}{2}+\cancel{x^2\ln y}=C\rightarrow\boxed{x^2\ln y-\cos x+\frac{(\ln y)^2}{2}=C}$$

d) $y'+y=xe^{6x}y^5$. **(3 points)**

$$\boxed{P(x)=1\quad Q(x)=xe^{6x}\qquad n=5}\rightarrow\boxed{P^*(x)=-4\qquad Q^*(x)=-4xe^{6x}\qquad 1-n=-4}$$

$$\mu=e^{\int-4dx}=e^{-4x}$$

$$y^{-4}=e^{4x}\left[\int-4xe^{2x}+C\right]$$

$$\boxed{y^{-4}=e^{4x}\left[-2xe^{2x}+e^{2x}+C\right]}$$

$-4x\searrow^{+}$	e^{2x}
$-4\searrow^{-}$	$\frac{1}{2}e^{2x}$
0	$\frac{1}{4}e^{2x}$

Question 2: [LO's: LO 1, LO 2, LO 4] **(10 points)**

Find the general solution of the following differential equations:

a) $\frac{d^5y}{dx^5}-3\frac{d^4y}{dx^4}-4\frac{d^3y}{dx^3}=0.$ **(3 points)**

$m^5-3m^4-4m^3=0.$

$m^3\left(m^2-3m-4\right)=0$

$m^3=0 \quad (m-4)(m+1)=0$

$m_1=m_2=m_3=0 \quad m_4=4 \;\; m_5=-1$

$$\boxed{y=C_1+C_2x+C_3x^2+C_4e^{4x}+C_5e^{-x}}$$

b) $y'''+4y''+14y'+20y=0.$ **(3 points)**

$m^3+4m^2+14m+20=0$

$m_1=-2 \qquad m_{2,3}=-1\pm 3i$

$$\boxed{y=C_1e^{-2x}+e^{-x}\left(C_2\cos 3x+C_3\sin 3x\right)}$$

c) $x^2y''+xy'-y=x$, given that $y=x$ is a homogenous solution. **(4 points)**

$y=xu\rightarrow y'=xu'+u\rightarrow y''=xu''+2u'$

$x^2\left(xu''+2u'\right)+x\left(xu'+u\right)-xu=x$

$x^3u''+2x^2u'+x^2u'+\cancel{xu}-\cancel{xu}=x$

$x^3u''+3x^2u'=x\rightarrow u''+\frac{3}{x}u'=\frac{1}{x^2}$

$z=u' \qquad z'=u''$

$z'+\frac{3}{x}z=\frac{1}{x^2}$

$\mu=e^{\int\frac{3}{x}dx}=e^{3\ln x}=e^{\ln x^3}=x^3$

$z=\frac{1}{x^3}\left[\int x\,dx\;+C_1\right]=\frac{1}{x^3}\left[\frac{x^2}{2}+C_1\right]$

$u'=z=\frac{1}{2x}+C_1x^{-3}$

$u=\frac{1}{2}\ln|x|-\frac{1}{2}C_1x^{-2}+C_2$

$$\boxed{y=\frac{x}{2}\ln|x|-\frac{1}{2}\frac{C_1}{x}+C_2x}$$

GOOD LUCK